

References

- Mathematics for Deep Learning

https://www.d2l.ai/chapter_appendix-mathematics-for-deep-learning/index.html

- Essential Mathematics for Machine Learning

(By Prof. Sanjeev Kumar, Prof. S. K. Gupta | IIT Roorkee | NPTEL)

https://onlinecourses.nptel.ac.in/noc21_ma38/preview

- Linear algebra for Data Science

(Shaina Race Bennett)

<https://www.youtube.com/playlist?list=PLB3yPBd26tWyDNoUpEGVsyl-sygPLqYa1>

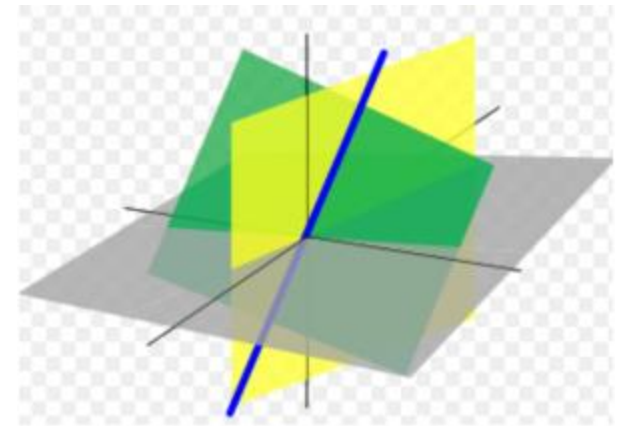
1. Linear Algebra

- Linear algebra is the branch of mathematics concerning linear equations such as

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

- By default, Vectors are denoted as Column Vectors.
- In vector notation we say $\mathbf{a}^T \mathbf{x} = b$

$$[a_1, a_2, \dots, a_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = b$$



- Called a linear transformation of $\mathbf{x}$
- Linear algebra is fundamental to geometry, for defining objects such as lines, planes, rotations.

2. Why Linear Algebra?

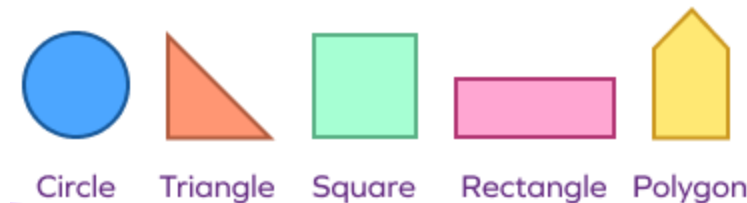
- Linear Algebra provides us with the mathematical tool to understand lower dimensions (2-D/3-D) and generalise for higher dimensions (n-D).

- 0-Dimensional : . (dot)



- 1-Dimensional

- 2-Dimensional

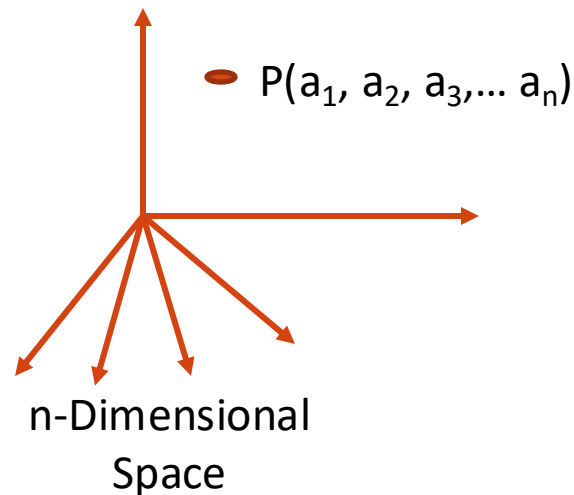
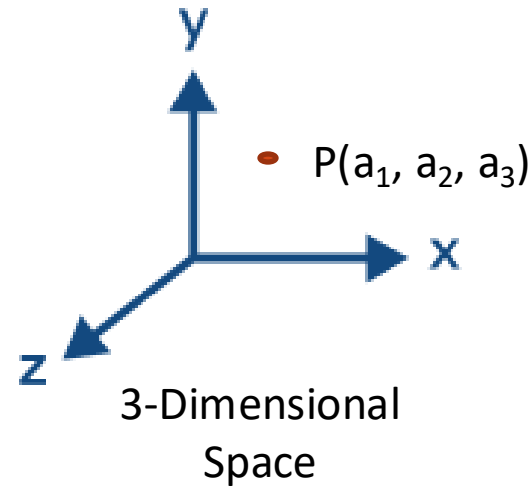
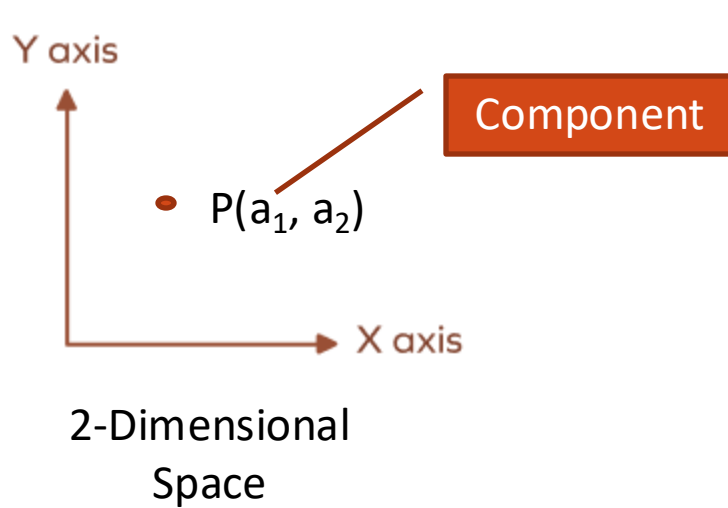


- 3-Dimensional

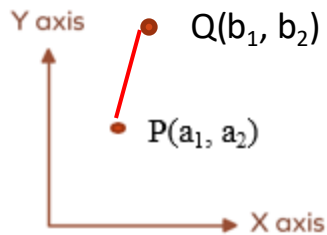


- n-Dimensional: Hypersphere, Hyperplane, Hypercube,...

3. Point (Vector)

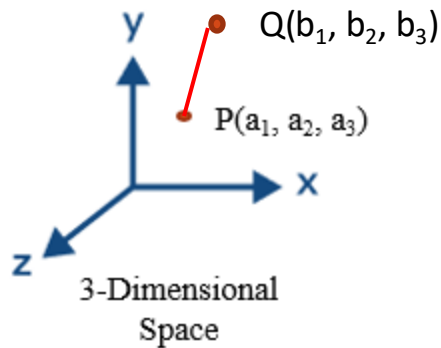


4. Distance between two Points

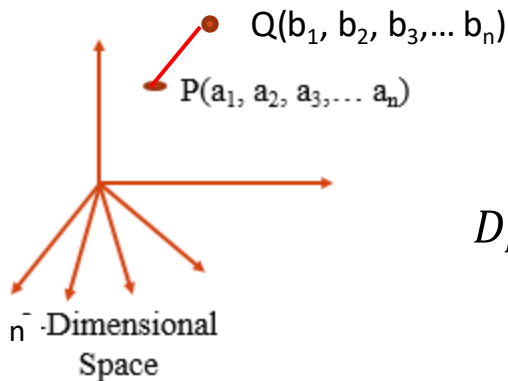


2-Dimensional
Space

$$D_{PQ} = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

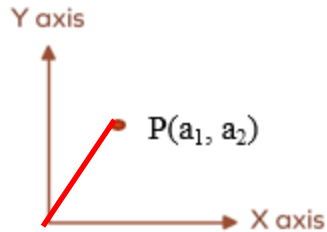


$$D_{PQ} = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2}$$



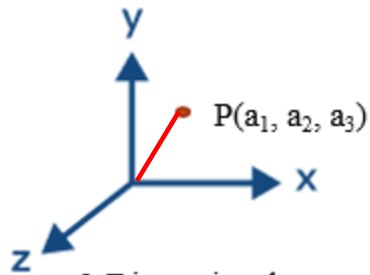
$$D_{PQ} = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2 + \dots + (a_n - b_n)^2} = \sqrt{\sum_{i=1}^n (a_i - b_i)^2}$$

4. Distance of a Point from Origin



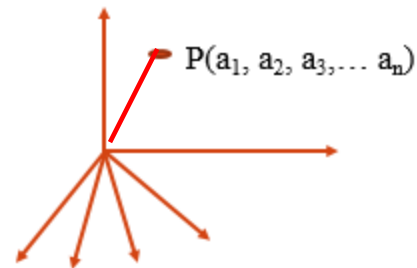
2-Dimensional
Space

$$D = |P| = \sqrt{(a_1 - 0)^2 + (a_2 - 0)^2} = \sqrt{a_1^2 + a_2^2}$$



3-Dimensional
Space

$$D = |P| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$



n-Dimensional
Space

$$D = |P| = \sqrt{a_1^2 + a_2^2 + a_3^2 + \dots + a_n^2} = \sqrt{\sum_{i=1}^n a_i^2}$$

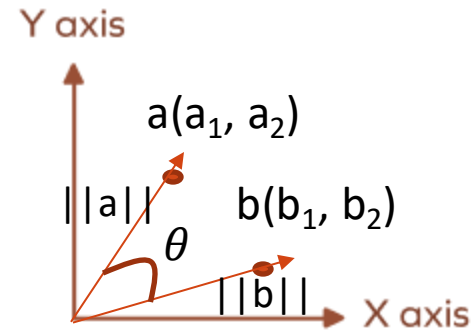
5. Vector Operations

- $a = [a_1, a_2, \dots, a_n]$
 - $b = [b_1, b_2, \dots, b_n]$
 - Addition: $a + b = [a_1 + b_1, a_2 + b_2, \dots, a_n + b_n]$
 - Subtraction: $a - b = [a_1 - b_1, a_2 - b_2, \dots, a_n - b_n]$
 - Multiplication:
 - Dot Product: $a \cdot b = [a_1 b_1 + a_2 b_2 + \dots, a_n b_n]$
- $$a \cdot b = [a_1, a_2, \dots, a_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = a^T b$$
- Cross Product (Not much used in Machine Learning)

5. Vector Operations

■ Dot Product: (Visualize in 2-D)

- $a \cdot b = \|a\| \|b\| \cos(\theta)$ [Geometry Definition]
- Where $\|a\| = \sqrt{a_1^2 + a_2^2}$ = distance of a from Origin



- $a \cdot b = a_1 b_1 + a_2 b_2$ [Algebra Definition]

- The angle between two vectors $= \theta = \cos^{-1} \left(\frac{a \cdot b}{\|a\| \|b\|} \right)$

$$\theta = \cos^{-1} \left(\frac{a_1 b_1 + a_2 b_2}{\|a\| \|b\|} \right)$$

When $\theta = 90^\circ \Rightarrow \cos(90) = 0 \Rightarrow a \cdot b = 0$

$$\begin{bmatrix} \vec{a} \\ 1 \\ 7 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 3 & \vec{b} & 7 \end{bmatrix} = 1 \cdot 3 + 7 \cdot 4 + 8 \cdot 7 = 3 + 28 + 56 = 87$$

Hadamard Product (Element-wise Multiplication)

- It is named after French Mathematician Jacques Hadamard.

$$\vec{g} \circ \vec{h} \circ \vec{m}$$

- The order of matrices/vectors to be multiplied should be the same, and the resulting matrix will also be of the same order.

$$\begin{matrix} & G & \\ \begin{bmatrix} 3 & 5 & 7 \\ 4 & 9 & 8 \end{bmatrix} & \circ & \begin{matrix} H \\ \begin{bmatrix} 1 & 6 & 3 \\ 0 & 2 & 9 \end{bmatrix} \end{matrix} = \begin{matrix} & N & \\ \begin{bmatrix} 3 \times 1 & 5 \times 6 & 7 \times 3 \\ 4 \times 0 & 9 \times 2 & 8 \times 9 \end{bmatrix} \end{matrix}$$

- Hadamard Product is used in LSTM (Long Short-Term Memory) cells of Recurrent Neural Networks (RNNs).

5. Vector Operations

- Dot Product: (In n-D)

- $a \cdot b = \|a\| \|b\| \cos(\theta)$

- Where $\|a\| = \sqrt{\sum_{i=1}^n a_i^2}$ = distance of a from Origin

- $a \cdot b = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^n a_i b_i$

- The angle between two vectors $= \theta = \cos^{-1} \left(\frac{a \cdot b}{\|a\| \|b\|} \right)$

$$\theta = \cos^{-1} \left(\frac{\sum_{i=1}^n a_i b_i}{\|a\| \|b\|} \right)$$

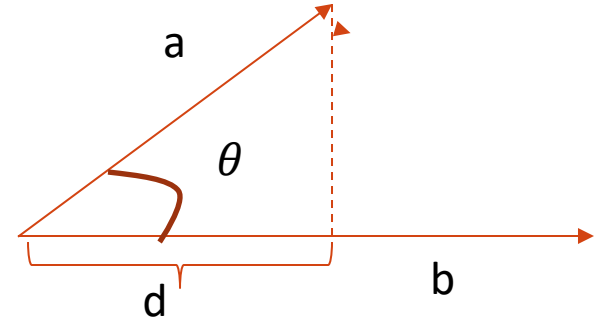
When $\theta = 90^\circ \rightarrow \cos(90) = 0 \rightarrow a \cdot b = 0$

$$a \cdot a = a_1 a_1 + a_2 a_2 + \dots + a_n a_n = \sum_{i=1}^n a_i^2 = \|a\|^2$$

Dot product between two same vectors = (distance from Origin)²

6. Projection

- $\cos(\theta) = \frac{p}{h} = \frac{d}{\|a\|}$
- *Projection of a on b i.e. $d = \|a\|\cos(\theta)$*
- $d = \frac{a \cdot b}{\|b\|} = \frac{\|a\|\|b\|\cos(\theta)}{\|b\|} = \|a\|\cos(\theta)$

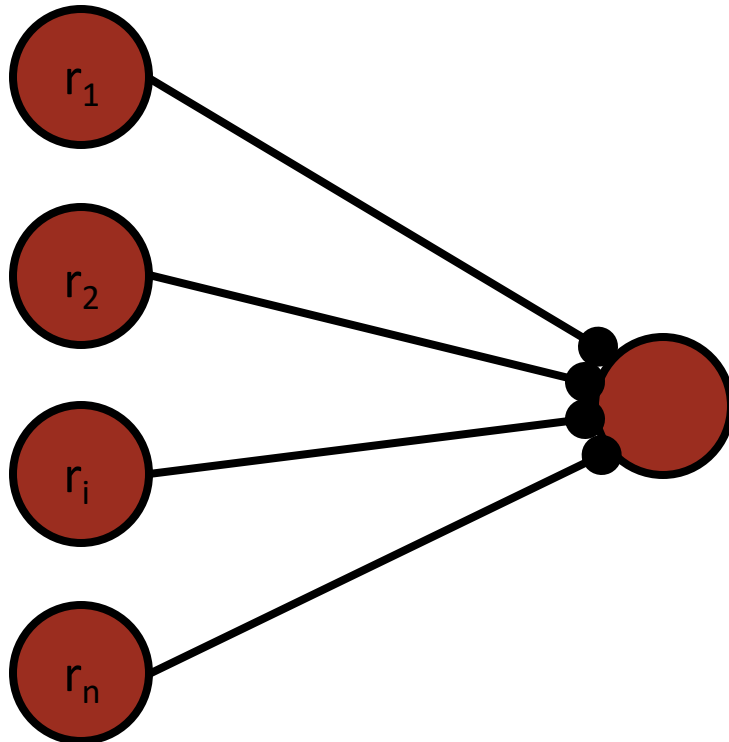


Example: linear feed-forward network

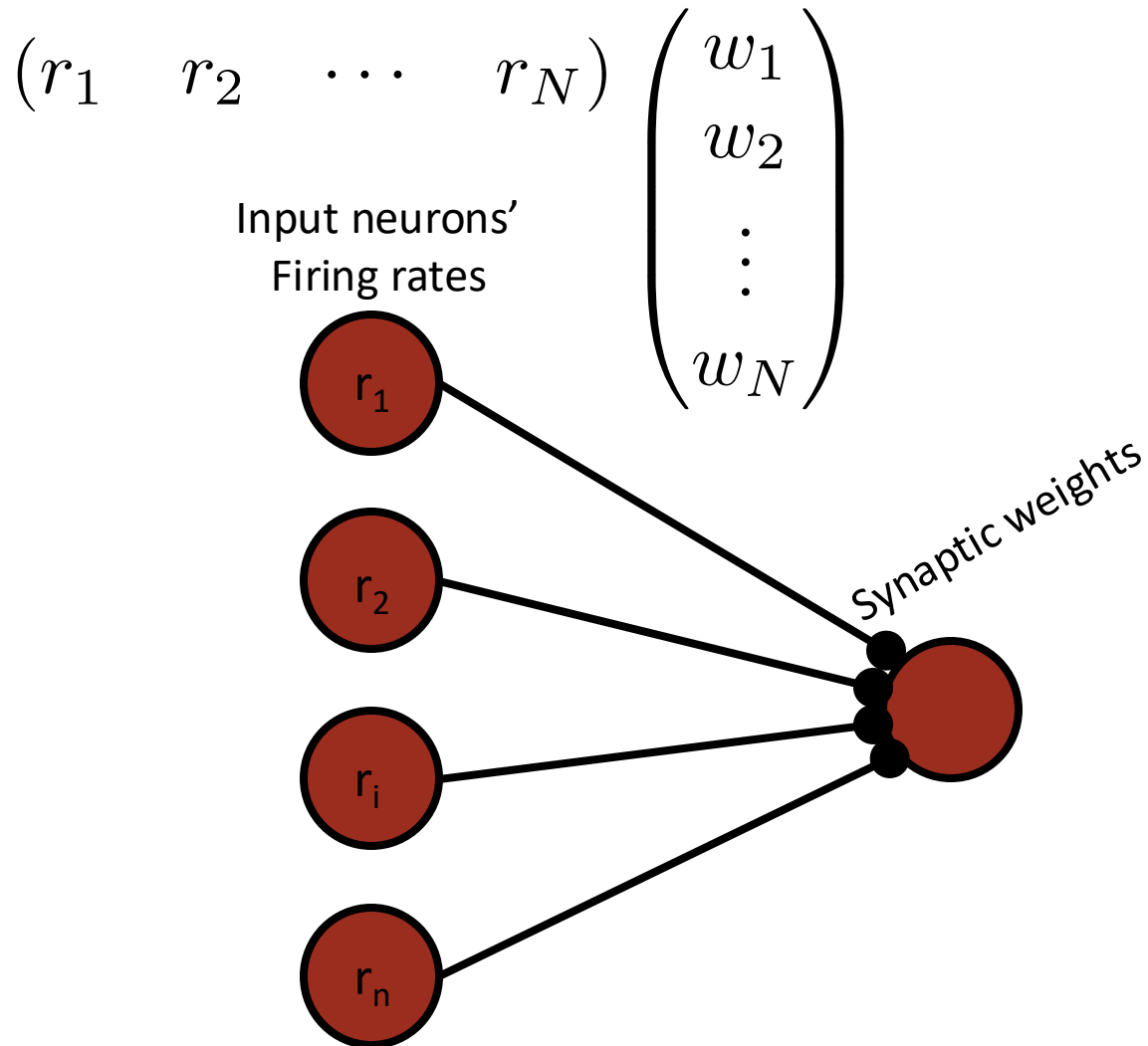


$$(r_1 \quad r_2 \quad \cdots \quad r_N)$$

Input neurons'
Firing rates



Example: linear feed-forward network

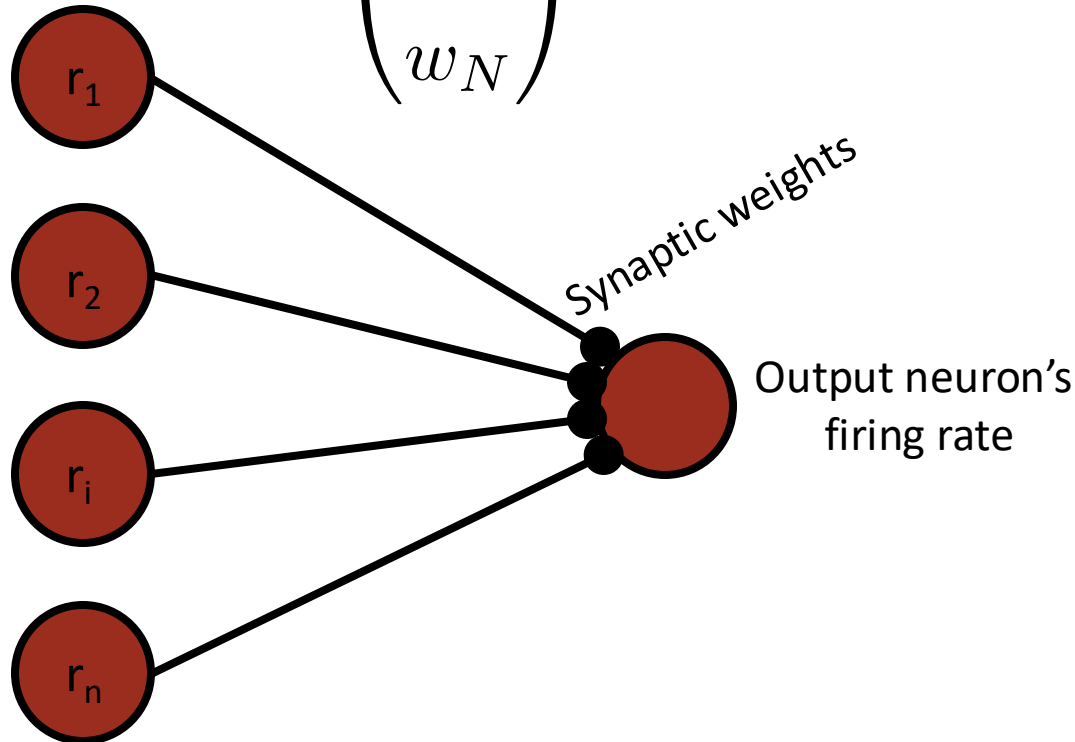


Example: linear feed-forward network



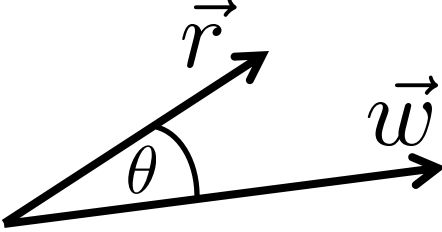
$$(r_1 \quad r_2 \quad \cdots \quad r_N) \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_N \end{pmatrix} = r_1 w_1 + r_2 w_2 + \cdots + r_N w_N$$

Input neurons' Firing rates



Example: linear feed-forward network




$$\vec{r} \cdot \vec{w} = |\vec{r}| |\vec{w}| \cos(\theta)$$

Input neurons'
Firing rates

r_1

r_2

r_i

r_n

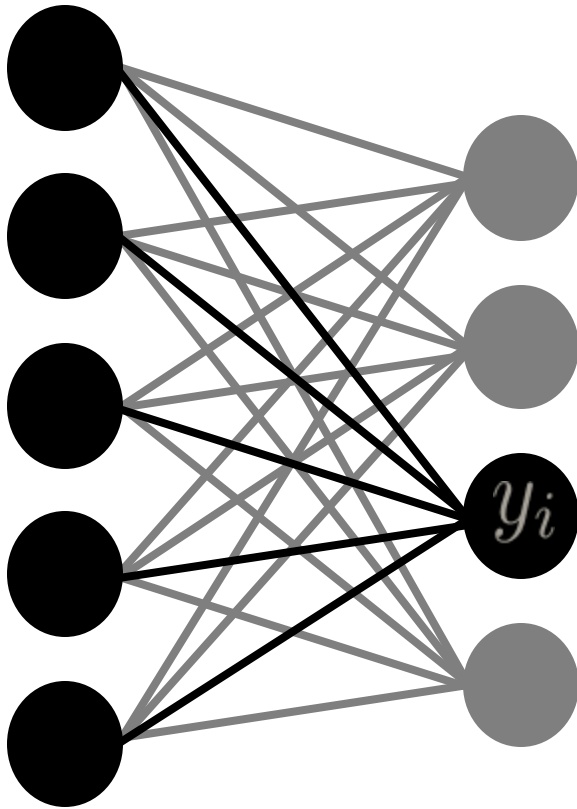
Synaptic weights

Output neuron's
firing rate

- Insight: for a given input (L2) magnitude, the response is maximized when the input is parallel to the weight vector
- Receptive fields also can be thought of this way

Example: 2-layer linear network: inner product point of view

- What is the response of cell y_i of the second layer?

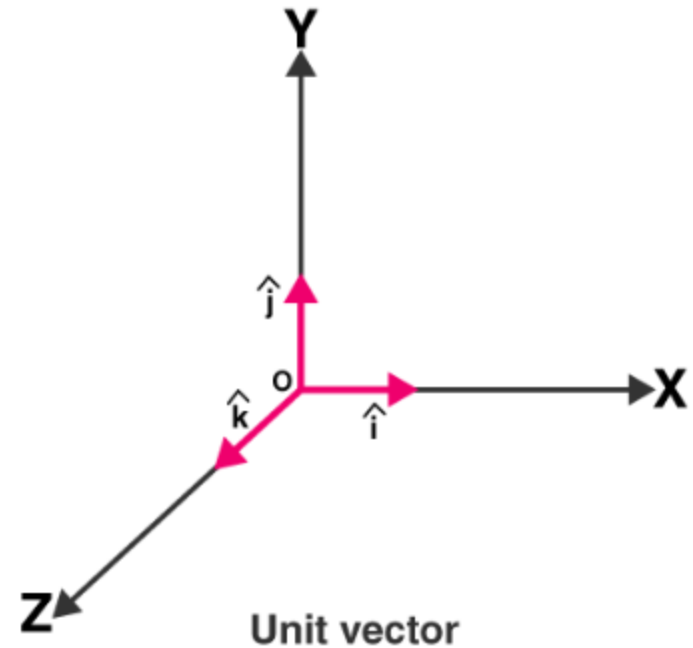


$$y_i = \sum_{j=1}^N W_{ij} x_j$$

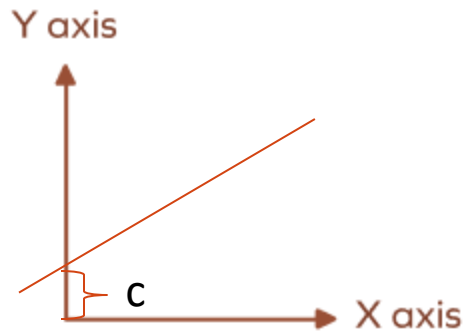
- The response is the dot product of the i^{th} row of W with the vector x

7. Unit Vector

- A vector is a quantity that has both magnitude, as well as direction.
- A vector that has a magnitude of 1 is a **unit vector**. It is also known as **Direction Vector**.
- Unit vector $\hat{a} = \frac{a}{\|a\|}$



8. Equation of a Line (2-D), Plane (3-D) & Hyperplane(n-D)

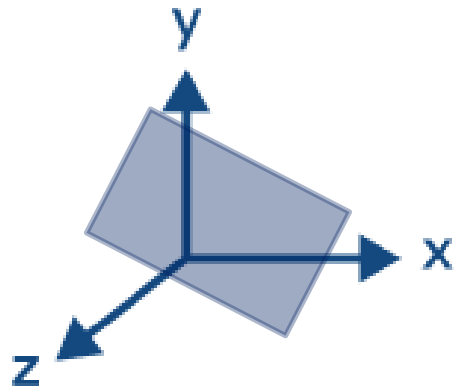


- 2-D:Line: $y = mx + c$

$$ax + by + c = 0$$

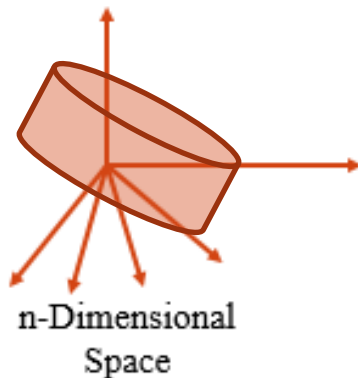
$$w_1x_1 + w_2x_2 + w_0 = 0$$

Note: When $w_0 = 0$, the line passes through the origin.



- 3-D: Plane:

$$w_1x_1 + w_2x_2 + w_3x_3 + w_0 = 0$$



- n-D:Hyperplane

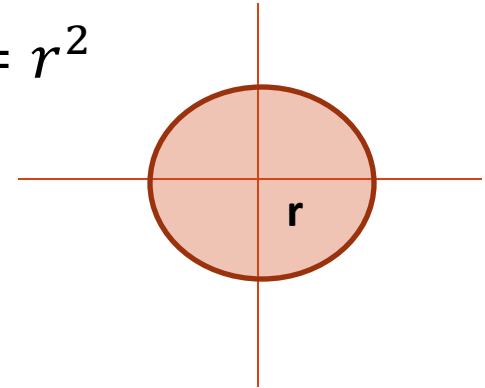
$$w_1x_1 + \dots + w_nx_n + w_0 = 0$$

$$w \cdot x + w_0 = 0$$

When $w_0 = 0$, the hyperplane passes through origin.

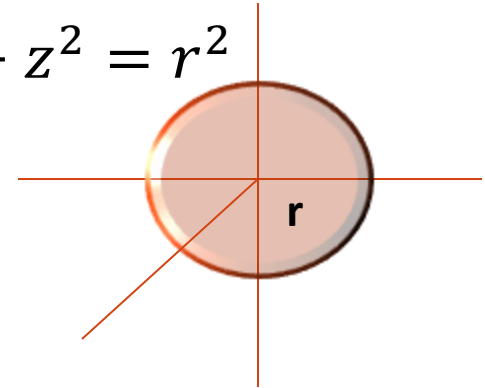
9. Equation of a Circle (2-D), Sphere (3-D) & Hypersphere(n-D)

- 2-D (Circle)
- Equation of a circle centred at origin: $x^2 + y^2 = r^2$
- A point $p(x_1, x_2)$ lies
 - Inside the circle if $x_1^2 + x_2^2 < r^2$
 - Outside the circle if $x_1^2 + x_2^2 > r^2$
 - On the circle if $x_1^2 + x_2^2 = r^2$



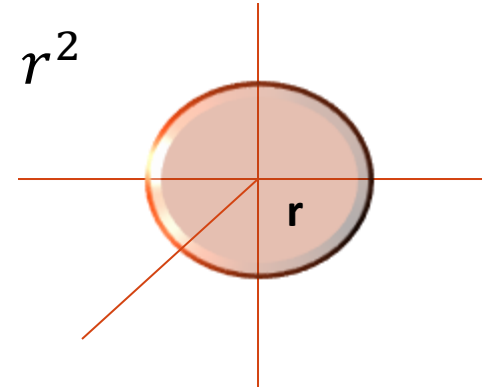
9. Equation of a Circle (2-D), Sphere (3-D) & Hypersphere(n-D)

- 3-D (Sphere)
- Equation of a circle centred at origin: $x^2 + y^2 + z^2 = r^2$
- A point $p(x_1, x_2, x_3)$ lies
 - Inside the circle if $x_1^2 + x_2^2 + x_3^2 < r^2$
 - Outside the circle if $x_1^2 + x_2^2 + x_3^2 > r^2$
 - On the circle if $x_1^2 + x_2^2 + x_3^2 = r^2$



9. Equation of a Circle (2-D), Sphere (3-D) & Hypersphere(n-D)

- n-D (Hypersphere)
- Equation of a circle centred at origin: $\sum_{i=1}^n x_i^2 = r^2$
- A point $p(x_1, x_2, x_3 \dots x_n)$ lies
 - Inside the circle if $\sum_{i=1}^n x_i^2 < r^2$
 - Outside the circle if $\sum_{i=1}^n x_i^2 > r^2$
 - On the circle if $\sum_{i=1}^n x_i^2 = r^2$



Other Structures

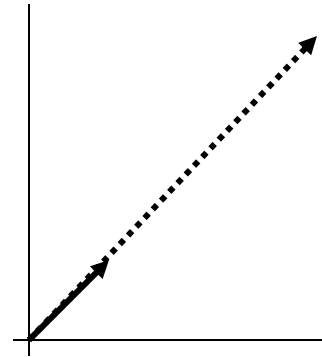
- Ellipse, Ellipsoid, Hyperellipsoid
- Square, Cube, Hypercube
- Rectangle, Hyperrectangle
- :

Eigenvectors & eigenvalues

Introduction to Linear Algebra
by Mark Goldman, and mily Mackevicius

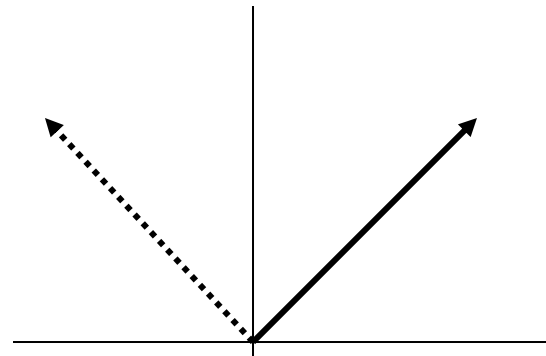
Matrices as linear transformations

$$\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$$



(stretching)

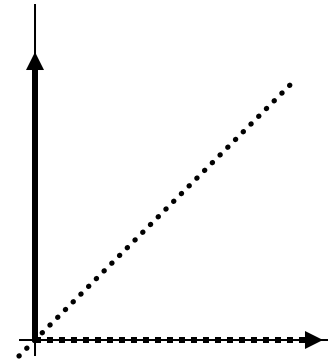
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$



(rotation)

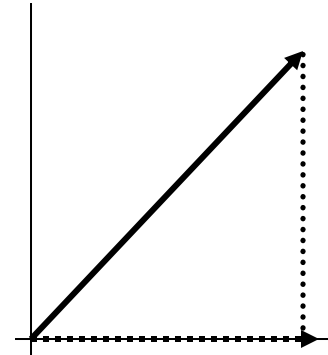
Matrices as linear transformations

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$



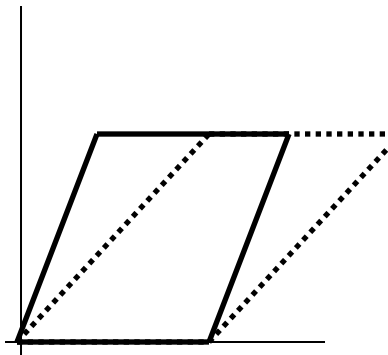
(reflection)

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$



(projection)

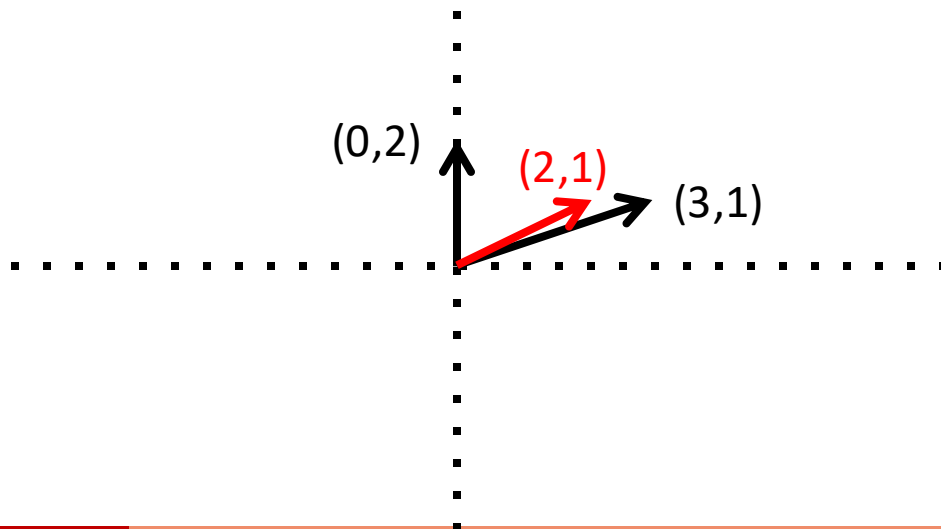
$$\begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + cy \\ y \end{pmatrix}$$



(shearing)

What do matrices do to vectors?

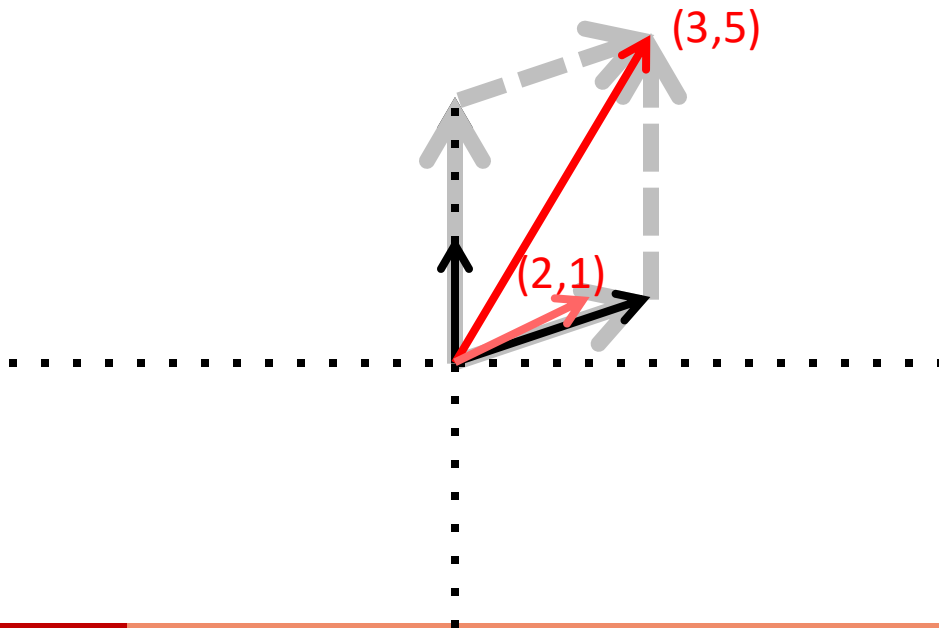
$$\begin{matrix} \overleftrightarrow{M} \\ \swarrow \\ \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \end{matrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} =$$



Recall

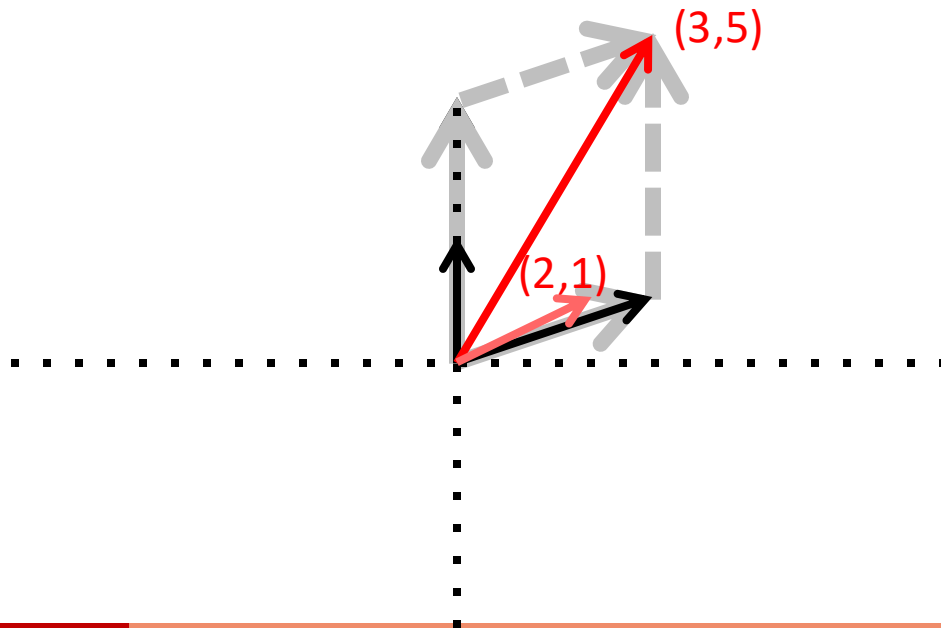
$\overrightarrow{M}$

$$\begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 2 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$



What do matrices do to vectors?

$$\begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 2 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$



- The new vector is:
 - rotated**
 - scaled**

Are there any special vectors that **only** get scaled?

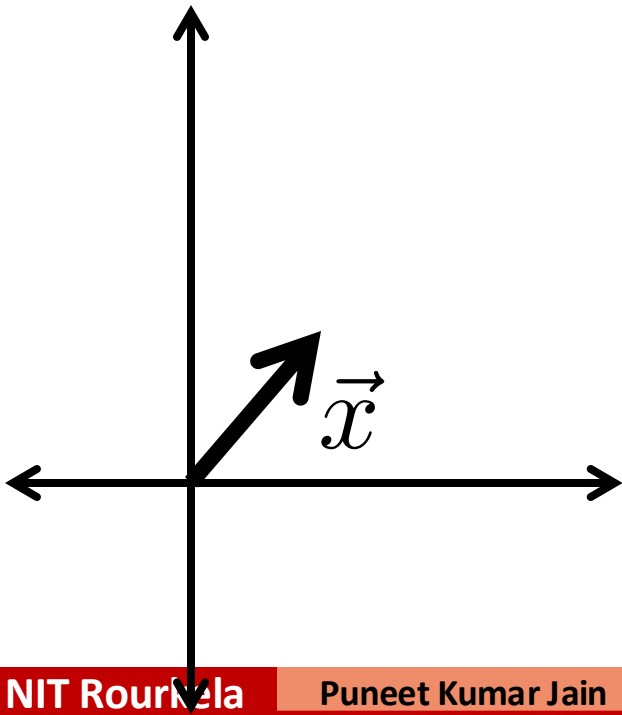


$$\vec{M} \rightarrow \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix}$$

Are there any special vectors that **only** get scaled?

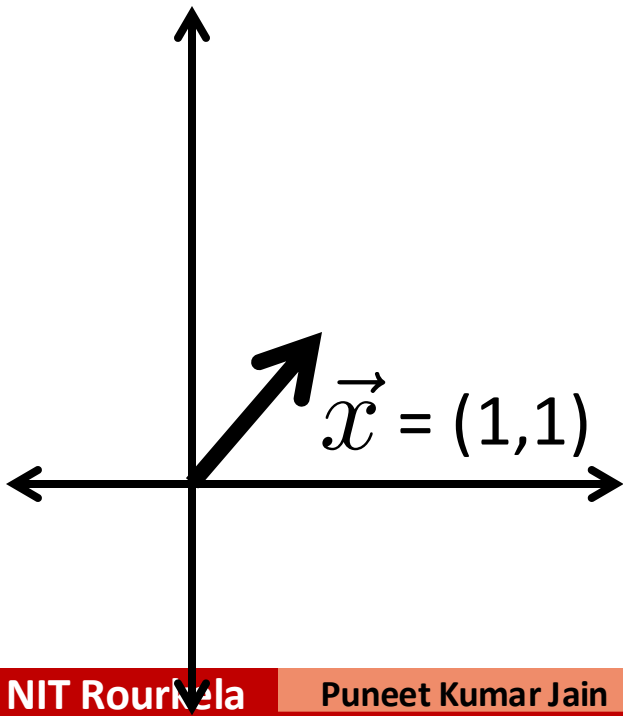
$$\vec{M} \rightarrow \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Try (1,1)



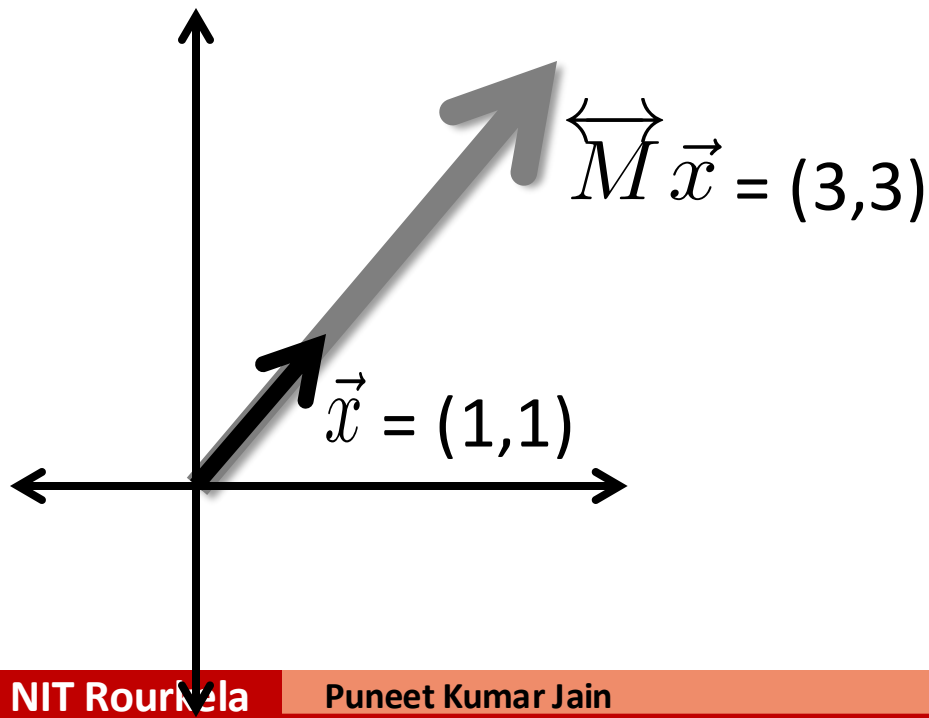
Are there any special vectors that **only** get scaled?

$$\vec{M} \rightarrow \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$$



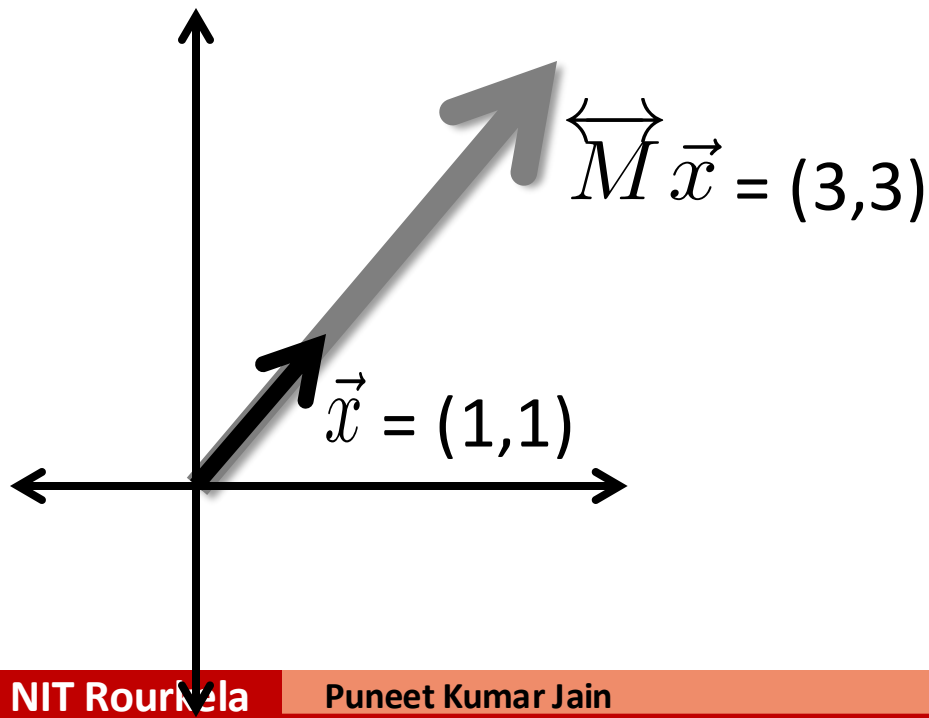
Are there any special vectors that **only** get scaled?

$$\begin{matrix} \overrightarrow{M} \\ \downarrow \end{matrix} \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$



Are there any special vectors that **only** get scaled?

$$\begin{matrix} \overleftrightarrow{M} \\ \downarrow \end{matrix} \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$



- For this special vector, multiplying by M is like multiplying by a scalar.
- $(1,1)$ is called an **eigenvector** of M
- 3 (the scaling factor) is called the **eigenvalue** associated with this eigenvector

- **Vector space**

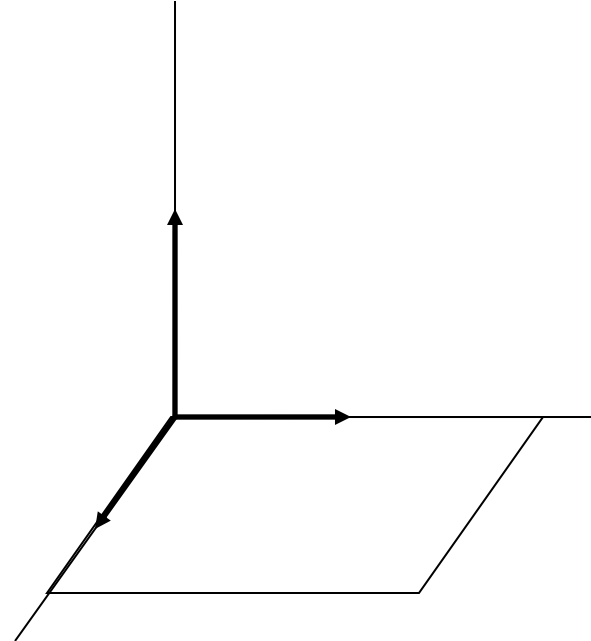
- 10-725 - Optimization
1/16/08 Recitation
- Joseph Bradley

Vector spaces

- Formally, a *vector space* is a set of vectors which is closed under addition and multiplication by real numbers.
- A *subspace* is a subset of a vector space which is a vector space itself, e.g. the plane $z=0$ is a subspace of $\mathbb{R}^3$ (It is essentially $\mathbb{R}^2$.).
- We'll be looking at $\mathbb{R}^n$ and subspaces of $\mathbb{R}^n$

Our notion of planes in $\mathbb{R}^3$ may be extended to *hyperplanes* in $\mathbb{R}^n$ (of dimension $n-1$)

Note: subspaces must include the origin (zero vector).

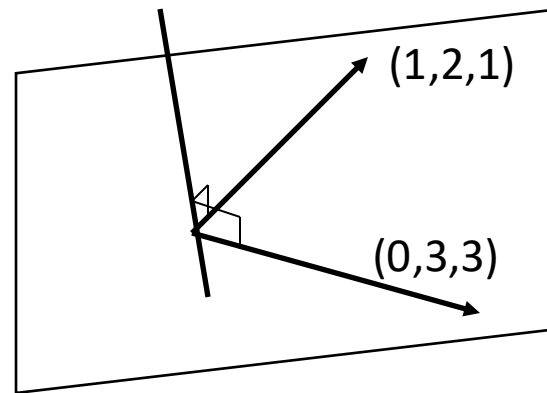


Linear system & subspaces

$$\begin{pmatrix} 1 & 0 \\ 2 & 3 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

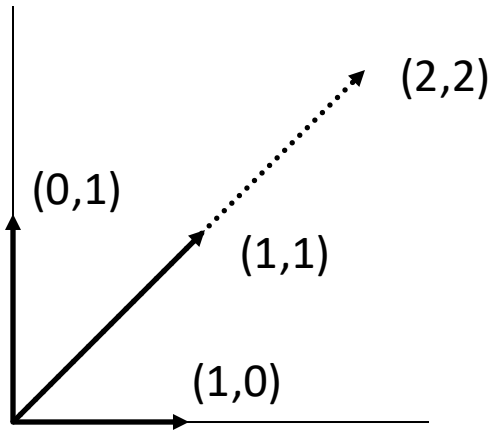
$$u \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + v \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

- Linear systems define certain subspaces
- $Ax = b$ is solvable iff b may be written as a linear combination of the columns of A
- The set of possible vectors b forms a subspace called the *column space* of A



Linear independence and basis

- Vectors $v_1, \dots, v_k$ are linearly independent if $c_1 v_1 + \dots + c_k v_k = 0$ implies $c_1 = \dots = c_k = 0$



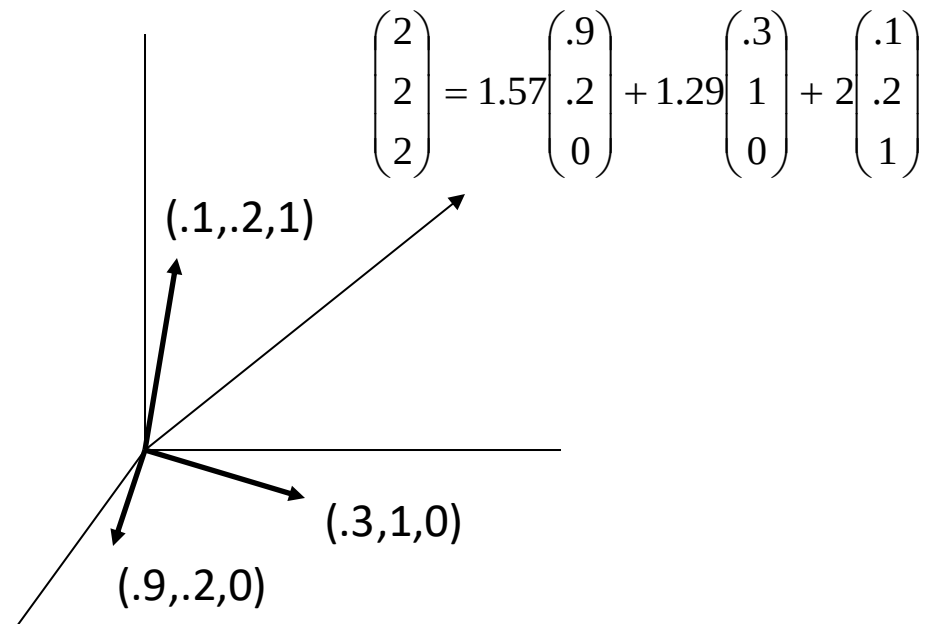
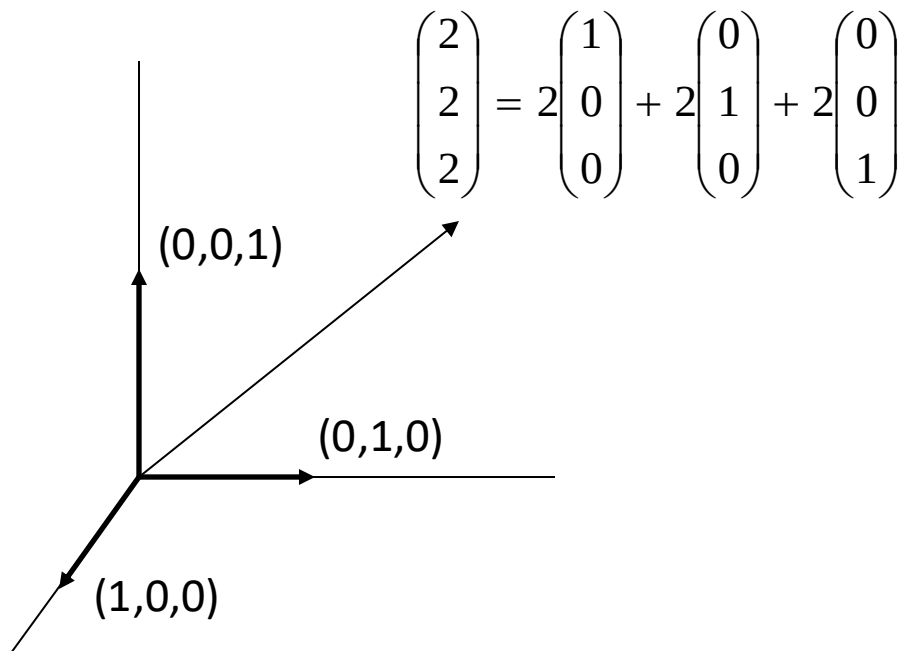
i.e. the nullspace is the origin

$$\begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

- If all vectors in a vector space may be expressed as linear combinations of $v_1, \dots, v_k$, then $v_1, \dots, v_k$ *span* the space.

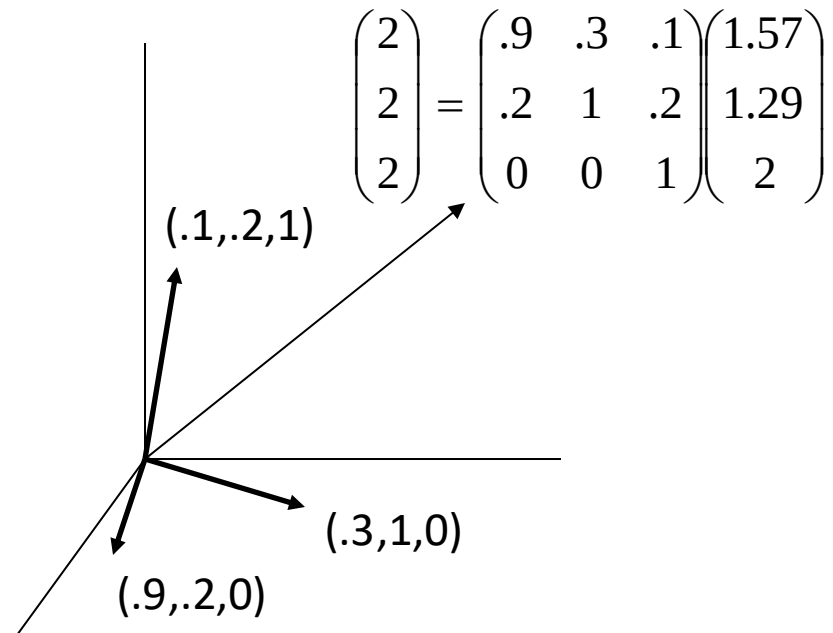
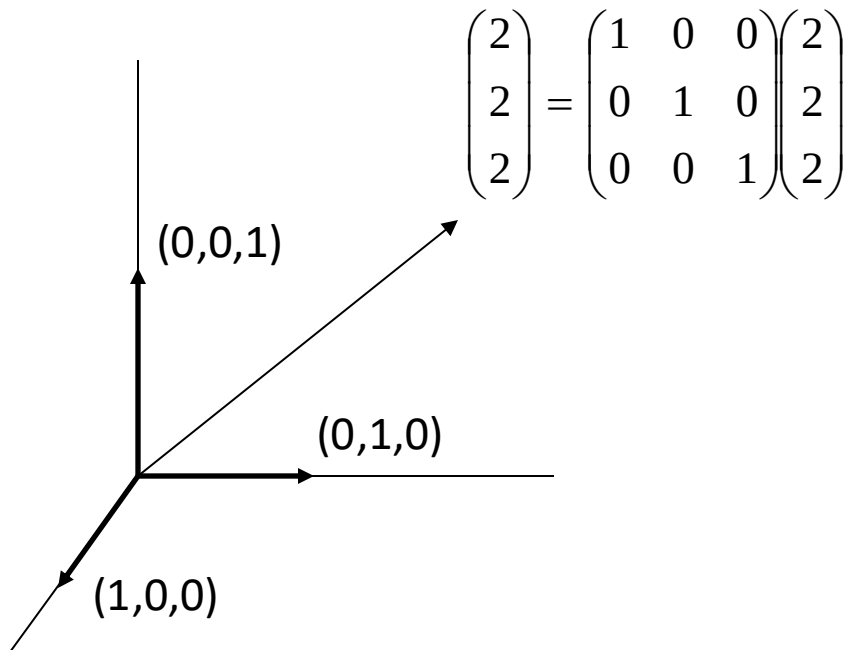
Linear independence and basis

- A *basis* is a set of linearly independent vectors which span the space.
- The *dimension* of a space is the # of “degrees of freedom” of the space; it is the number of vectors in any basis for the space.
- A basis is a maximal set of linearly independent vectors and a minimal set of spanning vectors.



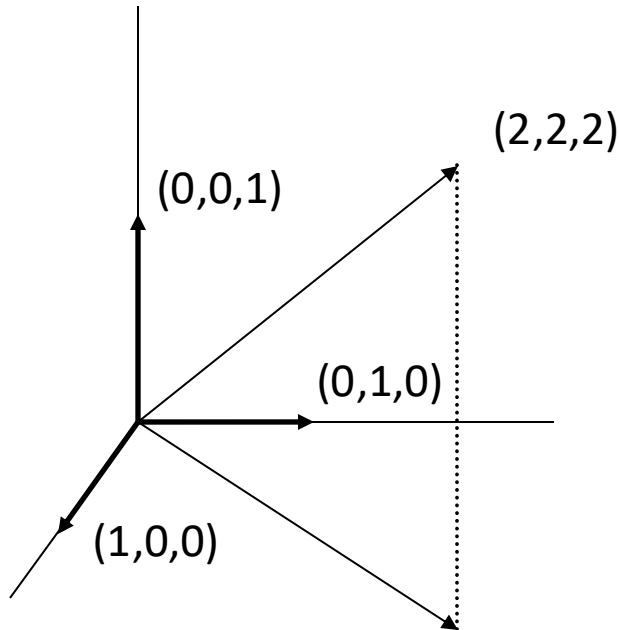
Basis transformations

We may write $v=(2,2,2)$ in terms of an alternate basis:

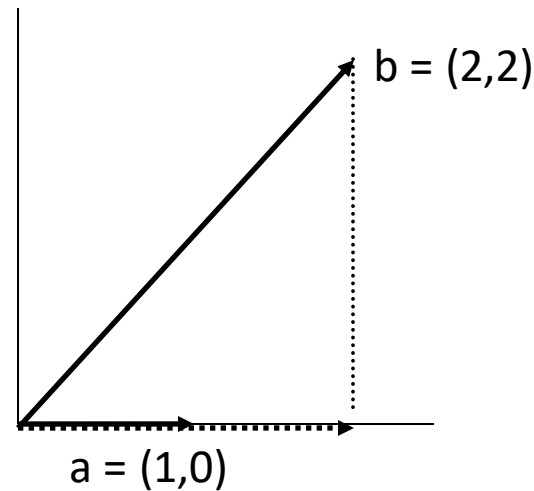


Components of $(1.57,1.29,2)$ are projections of v onto new basis vectors, normalized so new v still has same length.

Projections



$$\begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$$



$$c = \frac{a^T b}{a^T a} a = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

DFT as change of basis

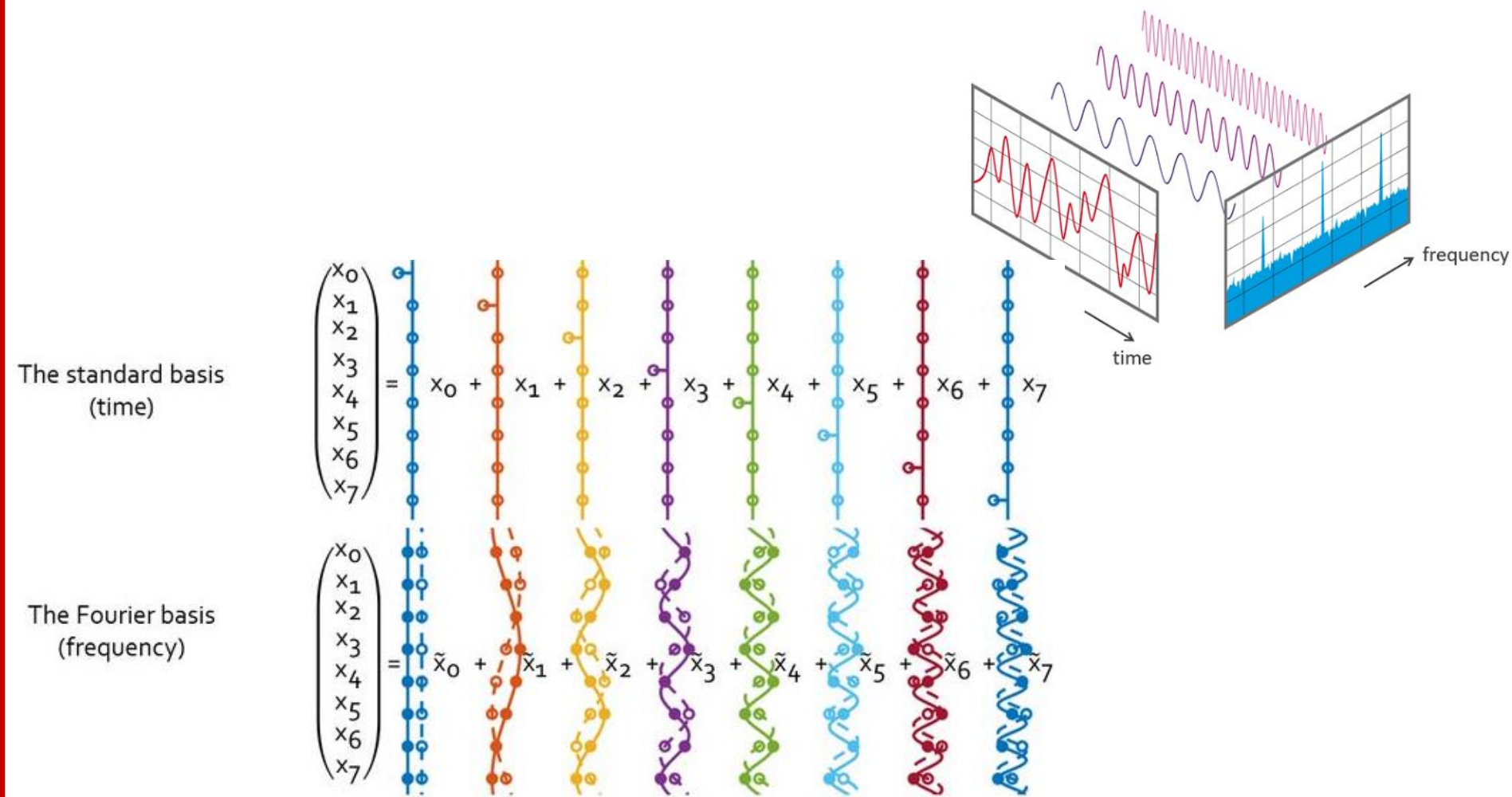


Image reference:

https://x.com/neuro_gal/status/1389539070129344514

About subspaces

- The *rank* of A is the dimension of the column space of A .
- It also equals the dimension of the *row space* of A (the subspace of vectors which may be written as linear combinations of the rows of A).

$$\begin{pmatrix} 1 & 0 \\ 2 & 3 \\ 1 & 3 \end{pmatrix}$$

$$(1,3) = (2,3) - (1,0)$$

Only 2 linearly independent rows, so rank = 2.

About subspaces

Fundamental Theorem of Linear Algebra:

If A is $m \times n$ with rank r ,

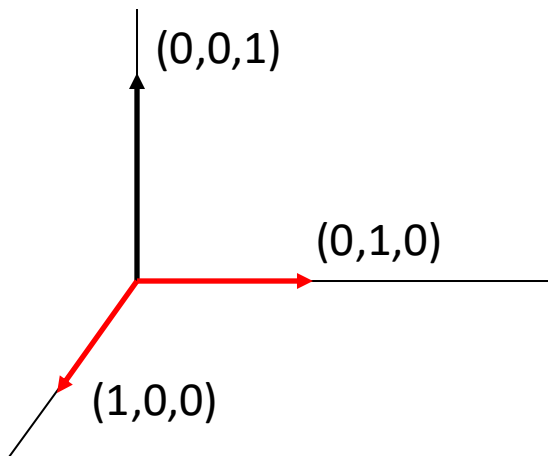
Column space(A) has dimension r

Nullspace(A) has dimension $n-r$ ($=$ *nullity* of A)

Row space(A) = Column space(A^T) has dimension r

Left nullspace(A) = Nullspace(A^T) has dimension $m - r$

Rank-Nullity Theorem: rank + nullity = n



$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$m = 3$$

$$n = 2$$

$$r = 2$$

Matrix inversion

- To solve $Ax=b$, we can write a closed-form solution if we can find a matrix A^{-1}
s.t. $AA^{-1}=A^{-1}A=I$ (identity matrix)
- Then $Ax=b$ iff $x=A^{-1}b$:
$$x = Ix = A^{-1}Ax = A^{-1}b$$
- A is *non-singular* iff A^{-1} exists iff $Ax=b$ has a unique solution.
- Note: If A^{-1} , B^{-1} exist, then $(AB)^{-1} = B^{-1}A^{-1}$,
and $(A^T)^{-1} = (A^{-1})^T$

Non-square matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 3 \end{pmatrix}$$

$$m = 3$$

$$n = 2$$

$$r = 2$$

System $Ax=b$ may not have a solution (x has 2 variables but 3 constraints).

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix}$$

$$m = 2$$

$$n = 3$$

$$r = 2$$

System $Ax=b$ is underdetermined (x has 3 variables and 2 constraints).

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Differentiation

Derivatives

⇒ Derivative →

Let's assume

$$y = x^2$$

$$\frac{dy}{dx} = 2x.$$

$$\frac{d^2y}{dx^2} = 2.$$

Partial derivative

$$y = \theta_1^2 + \theta_2^2$$

$$(\text{der}) - \frac{\partial y}{\partial \theta_1} = \begin{bmatrix} 2\theta_1 \end{bmatrix}$$

$$\frac{\partial y}{\partial \theta_2} = \begin{bmatrix} 2\theta_2 \end{bmatrix}$$

$\nabla_{\theta} L(\theta)$: vector of partial derivative with respect to θ_i

← This is called gradient

↓ Second order derivative.
(gradient of gradient)

$$\nabla_{\theta}^2 L(\theta) = \begin{bmatrix} \frac{\partial^2 L}{\partial \theta_1^2} & \frac{\partial^2 L}{\partial \theta_1 \partial \theta_2} \\ \frac{\partial^2 L}{\partial \theta_2 \partial \theta_1} & \frac{\partial^2 L}{\partial \theta_2^2} \end{bmatrix}$$

← Hessian matrix.

Derivatives

if $y = uv$

then

$$\frac{\partial y}{\partial x} = \frac{\partial u}{\partial x} \cdot v + u \frac{\partial v}{\partial x}$$

u and v are function of x and differentiable

$$\therefore u = x^2 \quad v = 3x$$

Ⓐ Product rule ↷

Ⓑ Quotient rule

$$\text{if } y = \frac{u}{v} = u \cdot v^{-1}$$

$$\frac{\partial y}{\partial x} = \frac{\partial u}{\partial x} \cdot v^{-1} + u \frac{\partial (v^{-1})}{\partial x} \rightarrow -\frac{1}{v^2} \frac{\partial v}{\partial x}$$

$$= \frac{\partial u}{\partial x} v^{-1} + \left(-\frac{u}{v^2}\right) \frac{\partial v}{\partial x}$$

$$= \frac{\frac{\partial u}{\partial x} v - u \frac{\partial v}{\partial x}}{v^2}$$

© Chain rule:

$$y = z^2 \quad \text{and} \quad z = 3x$$

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\partial z} \cdot \frac{\partial z}{\partial x}$$

another example:

$$z = e^{xy}$$

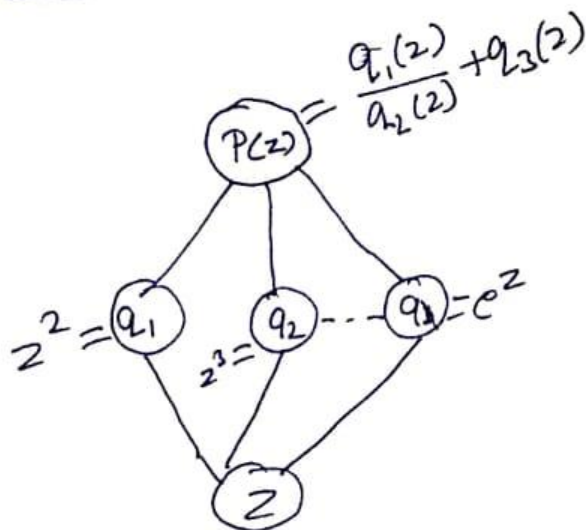
$$x = 2u + v$$

$$y = \frac{v}{v}$$

$$\frac{dz}{du} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\left| \begin{array}{l} \frac{\partial z}{\partial x} = y e^{xy} \\ \frac{\partial z}{\partial y} = x e^{xy} \\ \frac{\partial x}{\partial u} = 2 \\ \frac{\partial y}{\partial u} = \frac{1}{v} \end{array} \right.$$

generalization



$$\begin{aligned} \frac{dP(z)}{dz} &= \frac{\partial P(z)}{\partial q_1(z)} \cdot \frac{\partial q_1(z)}{\partial z} + \frac{\partial P(z)}{\partial q_2(z)} \cdot \frac{\partial q_2(z)}{\partial z} + \dots \\ &= \sum_{i=1}^n \frac{\partial P(z)}{\partial q_i(z)} \cdot \frac{\partial q_i(z)}{\partial z} \end{aligned}$$

End of Topic